Huawei Access Node QoS Congestion Event Resolution SOP (BAS-ACC-73)

Node: BAS-ACC-73 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_1827

Description

This Standard Operating Procedure (SOP) is designed to address QoS congestion events on Huawei Access nodes, specifically within the context of microbursts, interface tail drops, and QoS queue starvation. The failure domain is primarily attributed to the role of the Access node in the network architecture. In the Huawei Access node, QoS congestion events can arise due to the interaction of various factors, including policy-based routing, traffic policing, and shaping. The inherent hardware mechanics of the router's line cards and the associated ASICs (Application-Specific Integrated Circuits) play a crucial role in determining the overall QoS performance. The occurrence of microbursts can be attributed to the bursty nature of certain traffic patterns, which may overwhelm the interface's queue, leading to tail drops. Moreover, QoS queue starvation can result from inadequate configuration or misconfigured policies, causing certain traffic classes to be unfairly disadvantaged. Understanding these underlying protocol and hardware mechanics is essential for effective troubleshooting and resolution of QoS congestion events on the Huawei Access node.

Pre-requisite Checks

- check: Verify redundant link status and ensure it is operational., description: Confirm that the secondary link is up and functioning correctly.

- check: Dispatch remote hands to Basavanagudi POP for physical access., description: Ensure that authorized personnel are present at the POP to perform necessary physical checks and interventions.

- check: Review system logs for any recent software or configuration changes., description: Examine system logs for any recent updates, upgrades, or configuration modifications that may have contributed to the QoS congestion event.

Potential Root Causes

Cause	Probability
Policy-based routing misconfiguration	High
Inadequate traffic policing or shaping	Medium
Hardware-related issues with line cards or ASICs	Low

Diagnostic Commands & Expected Outputs

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Command: display interface brief, Expected Output: Interface
description, status, and traffic statistics., Description: This
command provides a brief overview of the interface, including its
description, status, and current traffic statistics. It helps in
identifying any interface-related issues that may be contributing to
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the QoS congestion event.

Command: display qos policy, Expected Output: QoS policy configuration and status., Description: This command displays the current QoS policy configuration and status, enabling the identification of any policy-related issues that may be causing the QoS congestion event.

Command: display traffic-statistics, Expected Output: Traffic statistics, including packet and byte counts., Description: This command provides detailed traffic statistics, including packet and byte counts, which can help in identifying any specific traffic patterns or classes that may be contributing to the QoS congestion event.
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Resolution Steps

Step 1: Step: Verify and correct policy-based routing configuration., Command: undo qos policy [policy-name], Description: Verify the policy-based routing configuration and correct any misconfigurations that may be contributing to the QoS congestion event. Use the undo qos policy command to remove any unnecessary policies and reconfigure the affected policies as necessary.

Step 2: Step: Adjust traffic policing or shaping., Command: traffic-statistics policing [policy-name] [policing-mode], Description: Adjust the traffic policing or shaping configuration to ensure that it is correctly applied to the affected traffic classes. Use the traffic-statistics policing command to configure or modify the policing or shaping parameters as necessary.

Step 3: Step: Replace or repair faulty hardware., Command: replace line-card [slot-number], Description: If hardware-related issues are identified, replace or repair the faulty hardware as necessary. Use the replace line-card command to remove the faulty line card and insert a new one, ensuring that the system remains operational throughout the process.

Post-Check Verification

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Command: display qos policy, Expected Output: QoS policy configuration and status., Description: Verify that the QoS policy configuration has been corrected and is functioning correctly. Use the display qos policy command to display the current QoS policy configuration and status.

Command: display traffic-statistics, Expected Output: Traffic statistics, including packet and byte counts., Description: Verify that the traffic policing or shaping configuration has been adjusted correctly and is functioning as intended. Use the display traffic-statistics command to display the current traffic statistics, including packet and byte counts.
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Roll-back Procedures

- Step: Restore original policy-based routing configuration., Command: undo qos policy [policy-name], Description: If the resolution steps fail to resolve the QoS congestion event, restore the original policy-based routing configuration using the undo qos policy command. This ensures that the system returns to its previous state, allowing further troubleshooting and resolution efforts.

- Step: Revert traffic policing or shaping configuration., Command: undo traffic-statistics policing [policy-name], Description: If the resolution steps fail to resolve the QoS congestion event, revert the traffic policing or shaping configuration to its original state using the undo traffic-statistics policing command. This ensures that the system returns to its previous state, allowing further troubleshooting and resolution efforts.